



GENSET NO LOAD – LOAD & PRE-TEST CHECKING REPORT

Date :	20 / JULY / 2017		
PI Model :	PI 313C	Type :	CLOSE
Prime Power :	284 KVA	Canopy Color :	ORANGE
Frq. :	60 Hz	P.F. _{LOAD}	1
Serial No :			
Job No :	8650		

CLIENT : SAUDI ARBIAN SERVICE SAR A/C

PROJECT :

GENSET DETAILS :

Engine :	CUMMINS	Model No :	QSL9 – G3	Serial No :	84507982
Alternator :	STAMFORD	Model No :	UCDI 1274K1	Serial No :	N17D166213
Controller :	DEEP SEA	Model No :	6120	Serial No :	6419711
C. Breaker :	METASOL	Amps	400 A	Model No :	ABN 803C
C. Transformer :	ELEMEX	Amps	800/5 A	Model No :	N/A
AVR :	STAMFORD	Model No :	SX460	Serial No :	111505 - 2229

GENSET RATING :

Testing Full Load Current :	345 Amps	Nominal Current :	432 Amps
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LOAD TEST WILL BE IMPLEMENTED AT AMBIENT CONDITIONS

Temperature :	42° C	Altitude :	
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LOAD TEST

LOAD	RUN TIME	FREQ	SPEED	VOLTAGE L-L			CURRENT			POWER	COOLANT TEMP.	OIL PRESSURE	CHARGING ALTERNATOR
				L1	L2	L3	L1	L2	L3				
%	Min	Hz	RPM	V	V	V	A	A	A	KW	°C	BAR	VDC
0%	5	60.0	1800	380	380	380	--	--	--	--	72	4.48	26.5
25%	5	60.0	1800	380	380	380	86	84	84	57	80	4.40	26.5
49%	5	60.0	1800	380	380	381	170	168	167	112	85	4.28	26.5
75%	5	60.0	1800	380	380	381	260	259	259	172	88	4.16	26.5
99%	10	60.0	1800	380	380	381	343	343	342	226	92	4.04	26.5
110%	10	60.0	1800	380	380	381	379	378	378	250	94	4.00	26.5

LOAD ACCEPTANCE TEST:

SUDDENLY BLOCK LOAD	VOLTAGE	FREQUENCY	CURRENT	RESPONSE TIME
%	V	Hz	A	SEC
49%	380	60.0	170	0
0%	392 – 380	61.5 – 60.0	0	5
50%	371 – 380	58.5 – 60.0	175	5

PROTECTION SYSTEM TEST:

OVER FREQUENCY	UNDER FREQUENCY	OVER VOLTAGE	UNDER VOLTAGE	OIL PRESSURE	COOLANT LEVEL	COOLANT TEMP
SD	SD	SD	SD	SD	SD	SD
67	53	460	340	1.0	OK	ECM 105

ACCESSORIES DETAILS :

Model	Serial No
Model	Serial No
Model	Serial No

Test Carried out by PI Supervisor :

Name : Eng. FARHAN SALEEM

Signature :

Date : 20 / JULY / 2017

Witnessed and approved by Client :

Name :

Signature :

Date :

Test Certificate

FOR

SALIENT POLE, SELF EXCITED / PMG EXCITED AND SELF REGULATED AC GENERATOR

Machine No. **N17D166213**

Type Test has been conducted on similar design machine
 STANDARDS: Generator generally conforms to BS 5000 : Part 3,
 Tested in accordance with BS EN 60034-1/IS/IEC 60034-1
 Rotor dynamically balanced to BS ISO 1940 - 1

ROUTINE TEST CERTIFICATE FOR BRUSHLESS AC GENERATOR:

FRAME: UCDI274K1	AVR: SX460	AVR NO: 1115052229	ENCLOSURE: IP23	
RATED KVA: 250	RATED VOLTS: 400	RATED AMPS: 360.8	FREQUENCY: 50	PHASE: 3
POWER FACTOR: .8	STR CONN: Series Star	STR WINDING: 311	DUTY: Base Continuous	WIRE: 4-Wire
EXC. VOLTS: 79	EXC.AMPS: 2.9	AMB TEMP: 40		

FOR 3 PHASE SEQUENCE LOOKING FROM DRIVE END FOR CLOCKWISE ROTATION :U-V-W

WINDING	COLD RESISTANCE IN Ohm(+/- 10 Milli Ohm) AT 20°C (MAX VALUES)	INSULATION RESISTANCE (M- Ohm)	HV TEST 2KV FOR 1 MIN	INSULATION CLASS
STATOR	0.0267	2+	OK	H
ROTOR	2.164	2+	OK	H

REGULATION TEST

FREQ.	VOLTS	AMPS	KVA	PF	KW	EXC.VOLT	EXC.AMP	% LOAD
52	401.35	0.00	0.00	0.00	0.00	12.3	.51	0.00
50	400	360.8	250	.8	200	79	2.9	100.00

REGULATION = (N.L.VOLTAGE-F.L.VOLTAGE / RATED VOLTAGE)*100 = 0.34 %

Item Description: AC Generator:UCD27.4.K.1.311.,1,,,14,,SX460.Black - RAL9011
 Remarks:

Date: 21-APR-17	Job No: 3481897-006	AUTHORISED BY Rameshwar B Karkhande Quality Assurance
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Certificate Of Conformity

Quality System Approval
 ISO 9001/2008

Frame Size : **UCDI274K1**Serial Number : **N17D166213**

Manufactured Date : 21-APR-17

KVA: **250**Authorized By Quality Assurance: **Rameshwar B Karkhande**

Cummins India Ltd.
Power Generation Business Unit

Engine Test Certificate

QSL9-G3

This is to certify that Engine Model

Serial Number ⁸⁴⁵⁰⁷⁹⁸²was tested using

Diesel as per ISO - 3046 Standard.

Corrected Full Load B.H.P to N.T.P conditions as per the

test standard is.....³⁴⁵ B.H.P.@.....¹⁵⁰⁰R.P.M.

399BHP @ 1800RPM

Date : 09-Jun-17 / /


Product Conformance